

Overview

COFFEE

Lots

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TEA

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Auctions

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Publishing

TokenHarvest API

Welcome to the TokenHarvest developer documentation. This API allows warehouse operators, commodity brokers, and exchange administrators to integrate with the digital agricultural commodity marketplace for Coffee and Tea in East Africa.

Base URL

All API requests should be prefixed with the following base URL:

```
https:// {your-domain} /api
```

Authentication

TokenHarvest uses Clerk for authentication. All endpoints require a valid Clerk Session Token passed in the Authorization header. Tokens can be obtained via the Clerk frontend SDK after a user signs in.

```
Authorization: Bearer <clerk_session_token>
```

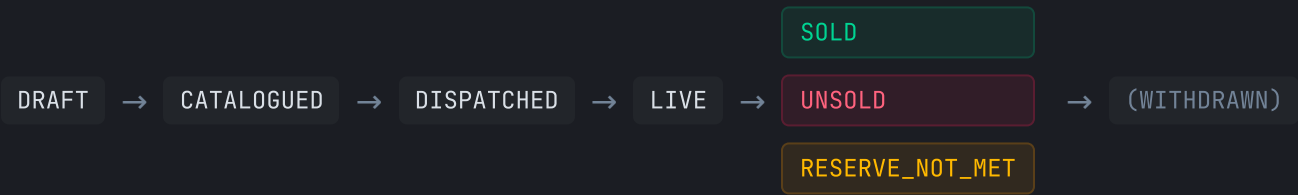
User Roles

Users are assigned specific roles which dictate their permissions within the marketplace.

Role	Description
ADMIN	Exchange administrator. Creates auction sessions, starts auctions, confirms payments.
ENABLER	Commodity broker. Catalogues lots (requires active mandate), submits lots to sessions.
PRODUCER	Warehouse receipt holder. Creates lots from their own eWRs directly. API-only role.

Lot Status Lifecycle

A lot progresses through several states from creation to final settlement.



Error Format

All endpoints return standardized JSON errors. The `issues` array is only present on 400 Validation Failed responses.

● ● ● JSON

```
{
  "error": "Human-readable message",
  "issues": [
    {
      "path": [
        "fieldName"
      ],
      "message": "Validation error detail"
    }
  ]
}
```

Coffee Lots

Manage the catalogue of coffee lots. Handle creation from electronic warehouse receipts (eWRs), updates, and dispatch documents.

List coffee lots

GET /api/coffee/lots

Retrieve a list of coffee lots. No authentication is required for browsing.

Query Parameters

Parameter	Type	Description
grade	string	Filter by coffee grade (e.g., AA, AB)
giOrigin	string	Filter by GI origin region
certification	string	Filter by certification (e.g., RainforestAlliance)
listingType	string	AUCTION or FIXED_PRICE
status	string	DRAFT, CATALOGUED, DISPATCHED, LIVE, SOLD, UNSOLD, WITHDRAWN, RESERVE_NOT_MET
brokerId	integer	Filter by mandate broker ID
ownerId	integer	Filter by eWR owner ID

Response Example

● ● ● JSON

```
[
  {
    "id": 42,
    "ewrId": 18,
    "ownerId": 7,
    "brokerId": 12,
    "grade": "AA",
    "gradeMark": "Gakuyu Farmers Cooperative",
    "giOrigin": "Nyeri",
    "grossWeightKg": "6120",
    "netWeightKg": "6000",
    "tareWeightKg": "120",
    "packageType": "60kg GrainPro Bags",
    "processingMethod": "Washed",
    "varietal": "SL28/SL34",
    "altitude": 1750,
    "cuppingRemarks": "Bright acidity, blackcurrant, bergamot. Score 87.",
    "certifications": [
      "RainforestAlliance",
      "FairTrade"
    ],
    "listingType": "AUCTION",
    "reservePriceUsd": "9.80",
    "status": "CATALOGUED",
    "warehouseCode": "NB0-001",
    "coffeeBeanSize": "17/18",
    "coffeeCuppingScore": "87",
    "publishedAt": "2026-08-07T09:00:00.000Z"
  }
]
```

Catalogue a new coffee lot

POST /api/coffee/lots

REQUIRES ROLE: ENABLER with active COFFEE mandate, or PRODUCER owning the eWR.

Request Body Schema

Field	Type	Required	Description
ewrId	integer	Yes	COFFEE eWR ID. Must be in INGESTED state.
grade	string	Yes	AA, AB, C, PB, E, TT, T, MH/ML
gradeMark	string	Yes	Factory or cooperative name
giOrigin	string	Yes	GI-registered region (e.g., Nyeri, Kirinyaga)
grossWeightKg	number	Yes	Total gross weight including packaging
netWeightKg	number	Yes	Net coffee weight after tare
tareWeightKg	number	Yes	Weight of packaging
packageType	string	Yes	e.g., 60kg GrainPro Bags
processingMethod	string	No	Washed, Natural, Honey, Wet-hulled
varietal	string	No	e.g., SL28/SL34, Bourbon, Geisha
altitude	number	No	Growing altitude in masl
cuppingRemarks	string	No	Tasting notes and subjective scores
certifications	string[]	No	RainforestAlliance, FairTrade, Organic, UTZ
listingType	string	No	AUCTION or FIXED_PRICE (default: AUCTION)
reservePriceUsd	number	Conditional	Required if listingType=AUCTION
fixedPricePerKgUsd	number	Conditional	Required if listingType=FIXED_PRICE
commissionRate	number	No	0.0 to 1.0 (default 0.01). Overridden by mandate.
bidSecurityPct	number	No	Bid deposit fraction, 0.0 to 1.0 (default 0.10)

Errors

- 400 Bad Request: Validation failed or eWR not in INGESTED state.
- 403 Forbidden: Caller does not have a mandate or own the eWR.
- 404 Not Found: eWR ID not found.
- 409 Conflict: An active lot already exists for this eWR.

Get full lot details

GET /api/coffee/lots/:id

Retrieves the complete profile for a single lot, including enriched data from the warehouse receipt and warehouse operator profile.

Response Example

● ● ● JSON

```
{
  "id": 42,
  "grade": "AA",
  "status": "CATALOGUED",
  "ownerName": "Jane Doe Farm",
  "brokerName": "Nairobi Exchange Brokers",
  "ewr": {
    "ewrsReceiptId": "EWR-COF-9912",
    "commodityType": "COFFEE",
    "weightMt": 6.12,
    "harvestSeason": "2025/2026",
    "state": "INGESTED",
    "coffeeBeanSize": "17/18",
    "coffeeCuppingScore": "87",
    "warehouseCode": "NB0-001"
  },
  "warehouseProfile": {
    "operatorName": "Kenya Warehouse Co.",
    "wrsLicenseNumber": "LIC-WH-992",
    "facilityType": "Bonded",
    "capacityMt": 5000,
    "warehouseInChargeName": "John Smith",
    "handlesCoffee": true
  }
}
```

Update a lot

PATCH /api/coffee/lots/:id

REQUIRES ROLE: Mandate broker or eWR owner.

Update fields on a DRAFT or CATALOGUED lot. Updating a DRAFT lot automatically transitions it to CATALOGUED.

All fields from the POST request are accepted, except `ewrId`.

Attach a dispatch document

POST /api/coffee/lots/:id/dispatch

Request Body

● ● ● JSON

```
{
  "docType": "PRE_AUCTION_DISPATCH",
  "docData": {
    "driverName": "Kiprop",
    "truckPlate": "KCC 1236",
    "sealNumber": "S-99102"
  }
}
```

Supported docType values:

- PRE_AUCTION_DISPATCH: Farm to auction warehouse dispatch.
- WEIGHMENT_REPORT: Official weighing at receiving warehouse.
- DELIVERY_ORDER: Post-sale delivery authorization (only allowed on SOLD lots).

List all dispatch documents

GET /api/coffee/lots/:id/dispatch

Retrieves an array of all dispatch records associated with the lot.

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Catalogue lot

Get lot details

Update lot

Attach dispatch

List dispatches

Coffee Auctions

Schedule sessions, manage catalogues, and run live coffee auctions on the exchange.

List auction sessions

GET /api/coffee/auctions

Query Parameters

Parameter	Type	Description
status	string	Filter by SCHEDULED, LIVE, CLOSED, or COMPLETED

Response Example

```
● ● ● JSON

[
  {
    "id": 3,
    "auctionDate": "2026-08-15",
    "scheduledStartTime": "09:00",
    "status": "SCHEDULED",
    "catalogueOrder": [
      42,
      43,
      44
    ],
    "createdByBrokerId": 12
  }
]
```

Create an auction session

POST /api/coffee/auctions

REQUIRES ROLE: ADMIN only.

Request Body Schema

Field	Type	Required	Description
auctionDate	string	Yes	Format: YYYY-MM-DD
startTime	string	No	Format: HH:MM (24-hour time)

Response

Returns a CoffeeAuctionSession object with status SCHEDULED.

Submit lots to session

POST /api/coffee/auctions/:sessionId/lots

REQUIRES ROLE: Mandate broker or ADMIN.

The target session must be in the SCHEDULED state. All submitted lot IDs must correspond to COFFEE commodity lots in CATALOGUED or DISPATCHED status.

Request Body

```
● ● ● JSON

{
  "lotIds": [
    42,
    43,
    44
  ]
}
```

Errors

- 400 Bad Request: Session is not SCHEDULED, or lots are ineligible.
- 403 Forbidden: Caller does not have rights to submit these lots.

Start a live auction

POST /api/coffee/auctions/:sessionId/start

REQUIRES ROLE: ADMIN only.

Moves a session from SCHEDULED to LIVE and begins the countdown for the first lot.

Request Body

```
● ● ● JSON

{
  "durationMins": 7
}
```

Response Example

```
● ● ● JSON

{
  "sessionId": 3,
  "currentLotId": 42,
  "auctionEndAt": "2026-08-15T09:07:00.000Z"
}
```

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Tea Lots

Manage the catalogue of tea lots. Tea lots share an identical lifecycle and structure to coffee lots, with a minor data model substitution for sensory evaluation.

Model Substitution

Instead of `processingMethod`, `varietal`, `altitude`, and `cuppingRemarks`, tea lots use a single field:

`tasterRemarks (string)`

Used for liquor analysis, appearance notes, and overall taster evaluation.

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List tea lots

`GET` `/api/tea/lots`

Supports the exact same query parameters as coffee lot listing.

Catalogue a new tea lot

`POST` `/api/tea/lots`

REQUIRES ROLE: `ENABLER` with active TEA mandate, or `PRODUCER` owning the TEA eWR.

Request Body Example

● ● ● JSON

```
{
  "ewrId": 9,
  "grade": "BOPF",
  "gradeMark": "Kericho Gold Estate",
  "giOrigin": "Kericho",
  "grossWeightKg": 3120,
  "netWeightKg": 3000,
  "tareWeightKg": 120,
  "packageType": "50kg Paper Sacks",
  "tasterRemarks": "Bright liquor, medium body, floral.",
  "certifications": [
    "RainforestAlliance"
  ],
  "listingType": "AUCTION",
  "reservePriceUsd": 3.2,
  "commissionRate": 0.01,
  "bidSecurityPct": 0.1
}
```

Get full lot details

`GET` `/api/tea/lots/:id`

Retrieves the complete profile including the TEA eWR data and warehouse profile.

Update a lot

`PATCH` `/api/tea/lots/:id`

REQUIRES ROLE: `Mandate broker` or `eWR owner`.

Same rules apply as coffee lots.

Attach a dispatch document

`POST` `/api/tea/lots/:id/dispatch`

Accepts the exact same `docType` values and payload structure as coffee dispatch attachments.

List all dispatch documents

`GET` `/api/tea/lots/:id/dispatch`

Tea Auctions

Schedule sessions, manage catalogues, and run live tea auctions. The endpoints mirror the coffee auction system exactly, with payloads and responses retaining the identical structure.

List auction sessions

GET /api/tea/auctions

Query by **status** (SCHEDULED, LIVE, CLOSED, COMPLETED).

Create an auction session

POST /api/tea/auctions

REQUIRES ROLE: ADMIN only.

Requires **auctionDate** and optional **startTime** .

Submit lots to session

POST /api/tea/auctions/:sessionId/lots

REQUIRES ROLE: Mandate broker or ADMIN.

Pass an array of **lotIds** . Target lots must be TEA commodity.

Start a live auction

POST /api/tea/auctions/:sessionId/start

REQUIRES ROLE: ADMIN only.

Moves the session to LIVE state. Provide **durationMins** .

Marketplace Publishing

Push catalogued commodity lots directly to the external TokenHarvest discovery marketplace. These endpoints handle the synchronous triggers and asynchronous state updates of the publishing pipeline.

Publish a lot to marketplace

POST

/api/tea/lots/:id/publish

Triggers a publication workflow. This endpoint is concurrency-safe: DB-level deduplication ensures only one in-flight publication occurs per lot.

Eligibility: The lot must NOT be in DRAFT, SOLD, UNSOLD, WITHDRAWN, or RESERVE_NOT_MET status.

Response 201 Created (or 200 OK on re-publish)

● ● ● JSON

```
{
  "id": 1,
  "listingId": 42,
  "factoryId": 12,
  "marketplaceName": "tokenharvest",
  "status": "live",
  "externalListingId": "EXT-2026-0042",
  "lastAttemptAt": "2026-08-07T10:00:00.000Z",
  "lastSuccessAt": "2026-08-07T10:00:01.200Z",
  "lastError": null
}
```

Response 409 Conflict

Returned when the lot is already successfully published.

● ● ● JSON

```
{
  "error": "Lot is already live on the marketplace.",
  "publication": {
    "id": 1,
    "status": "live",
    "externalListingId": "EXT-2026-0042"
  }
}
```

Retry failed publication

POST

/api/tea/lots/:id/publish/retry

Attempts to re-publish a lot where the initial publication failed (e.g. adapter timeout, external API failure). Only valid when the current publication record is in the `failed` state.

Errors

- 409 Conflict: Publication is not in failed state.

List publication records

GET

/api/listing-publications

Retrieve status histories of external publications. Scoped strictly to lots the caller owns or brokers.

Query Parameters

Parameter	Type	Description
lotId	integer	Filter by internal lot ID (ownership verified)
factoryId	integer	Filter by factory (must match caller context)